

RDHIS2 Mobile

Hardware Specification & Deployment Requirements

Device requirements for field deployment · rationale · storage-growth model · procurement guidance

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Build figures read from the app configuration; sizing figures marked (estimate) are for a national-scale DHIS2 instance.

1 At a Glance

	Minimum	Recommended
Android OS	7.0 (API 24, Nougat)	11 (API 30) or newer
Architecture	32-bit ARM (armeabi-v7a)	64-bit ARM (arm64-v8a)
RAM	2 GB	3–4 GB
Free storage at install	1 GB	4 GB+ (device ≥ 32 GB)
Display	5.0", 720 × 1280 (HD)	6"+ phone, or 8–10" tablet, 1080 × 1920
Battery	2500 mAh	3500 mAh+
Connectivity	Any 3G / Wi-Fi, intermittent	4G / LTE or Wi-Fi for the first sync
iOS (alternative)	iOS 13.0 — iPhone 6s / iPad (5th gen)	iOS 15+ — iPhone 11 / iPad (8th gen)
Sensors / peripherals	None required	—

The app is portrait-only and does not use the camera, GPS, fingerprint reader, NFC, Bluetooth, SD card, or a SIM / cellular connection for daily work (see §6).

2 Android — Detailed

2.1 Operating system

Item	Value	Source / rationale
minSdkVersion	24 — Android 7.0 (Nougat)	Resolved from flutter.minSdkVersion (Flutter 3.41.4). The app will not install below this.
targetSdkVersion	36 — Android 16	Keeps the app current with Google Play's target-API policy and the scoped-storage / notification-permission / foreground-service-type behaviours it implements.
compileSdk	36	—
Java 8+ APIs on older devices	Enabled via core-library desugaring	flutter_local_notifications uses java.time; desugaring makes it work down to API 24.

Practically: any Android phone or tablet sold from 2017 onward meets the OS floor. Android 10+ is recommended so the OS's own background-execution and notification controls behave predictably.

2.2 CPU architecture

The release build ships native libraries for:

ABI	Status	Notes
arm64-v8a	Recommended	Every 64-bit ARM phone / tablet. Best performance.
armeabi-v7a	Supported	Older / low-end 32-bit ARM devices.
x86_64	Emulator / Chromebook only	Not a field target.

A per-ABI split APK is ~25–28 MB; the universal APK is ~71 MB. Distribute the split APK or an Android App Bundle so each device downloads only its own ABI.

The release build has no R8/ProGuard minification yet — enabling it would cut the APK and install size by roughly 30–40 %.

2.3 Memory (RAM)

	Value	Rationale
Minimum	2 GB	The Flutter runtime plus the app's working set is ~150–300 MB. The first metadata sync decodes a multi-megabyte JSON payload on a background isolate — the transient peak is the memory-critical moment. On 1 GB devices that sync risks a low-memory kill mid-download, leaving the device needing a retry.
Recommended	3–4 GB	Headroom for the metadata decode, fl_chart dashboard rendering, and PDF / Excel export of a large form, with the OS keeping the app resident between sessions.

2.4 Storage

Component	Size	Notes
App install (per-ABI)	~55–90 MB (estimate)	Download ~25–28 MB; extracted native libs + assets roughly double it. Universal APK install is ~150 MB.
Bundled assets	3.2 MB	Noto Sans Ethiopic font (1.1 MB, for Amharic period labels) + images.
Per-user database	~20–80 MB typical, up to ~150 MB (estimate)	One SQLite file per logged-in user (hisp_<user>.sqlite + WAL/SHM sidecars). Dominated by the instance-wide metadata cache. Grows ~2–5 MB per year of active entry, plus the local audit trail.
Caches	< 5 MB	Outlier-history snapshots and local-dashboard result caches (JSON in the DB).

Sizing rule: provision at least 1 GB free at install and budget ~200 MB per user who will log in on a shared device. A 16 GB device is the practical floor; 32 GB+ is comfortable. The app never writes to an SD card or shared storage.

2.5 Display

- Portrait orientation is locked.
- Minimum usable: 5.0", 720 × 1280 (HD).
- Recommended: a 6"+ phone, or better an 8–10" tablet. The data-entry form is a table (data elements as rows, category-option-combo disaggregations as columns) plus collapsible sections — more width means less horizontal scrolling per row and fewer entry errors. For facilities capturing large disaggregated datasets a tablet materially improves entry speed and accuracy.
- Rendering uses Flutter's Impeller engine (Vulkan, GLES 3 fallback) — satisfied by any device meeting the API-24 floor. No dedicated GPU class is required.

2.6 Battery & power

- A push in flight is protected by an Android foreground service (dataSync type) so the OS does not kill it when the app is backgrounded.
- Deadline reminders use inexact alarms (no SCHEDULE_EXACT_ALARM) — low power impact, but the device must be powered on around 08:00 local to receive them.
- Recommend ≥ 3500 mAh for a full field day of intermittent capture and sync, or a power bank / vehicle charger for mobile teams.

3 iOS (Alternative Platform)

Item	Value
Deployment target	iOS 13.0
Minimum device	iPhone 6s / iPhone SE (1st gen) / iPad (5th gen) / iPad mini 4
Recommended	iOS 15+, iPhone 11 or newer, iPad (8th gen) or newer
Background sync	Best-effort only — iOS caps background execution at a few seconds; no foreground-service equivalent. Sync completes reliably only while the app is open.
Notifications	Require the user to grant permission on first launch.

Storage, RAM, and display guidance match the Android figures in §2.

4 Web (Development & Demonstration Only)

The web build is not a field-deployment target — it exists for development and demos.

Item	Requirement
Browser	Chromium-based (Chrome / Edge) recommended; any browser with WebAssembly + a persistent store.
Persistence	OPFS preferred; falls back to IndexedDB. A private-browsing window will not persist the database.
Assets	sqlite3.wasm (732 KB) + drift_worker.js (344 KB) from web/, plus the Flutter web engine.
Host	Served from localhost:3000 or localhost:3001 against staging — its CORS policy allows no other origin. Use ./run_web.sh (pins port 3001).
Machine	Any modern desktop / laptop, ~4 GB RAM.

5 Connectivity

The app is offline-first: a health worker completes an entire day's or month's capture with no network and loses nothing. Connectivity is needed only to seed and drain the local database.

Phase	Requirement	Volume (estimate)
First login + full metadata sync	A stable connection for ~2–10 minutes. The one time a solid link matters — an interrupted first sync must be retried from the start.	~5–30 MB download
Returning login (delta sync)	Brief connectivity.	Typically < 1 MB
Data push	Brief connectivity; auto-retried on reconnect, on login, and on a 5-minute heartbeat.	Small — values batched at 500 per request
Daily capture	None.	—
Dashboards / analytics	Live connection to view (results then cached offline).	Varies per chart

- Any bearer works: 2G/3G/4G/LTE/5G mobile data, or Wi-Fi. No SIM is required if Wi-Fi is available for the periodic sync.
- Works behind HTTP proxies (id-list requests are chunked to keep URLs short).

- Transport is HTTPS only in release builds; the server's TLS certificate must chain to a CA in the device's system trust store (no certificate pinning).

6 Sensors & Peripherals — What Is NOT Required

Procurement can safely choose devices without any of the following. The app requests none of the corresponding permissions.

Capability	Required?
Rear / front camera	No
GPS / location	No
Fingerprint / face unlock	No (biometric app-lock is a roadmap item, not built)
NFC	No
Bluetooth	No
SD card / external storage	No
Cellular modem / SIM (for daily use)	No — only for the periodic sync, and Wi-Fi substitutes
Google Play Services / GMS	No — no Firebase, Maps, or Play-only APIs
Accelerometer / gyroscope / other sensors	No

The only Android permissions requested are INTERNET, ACCESS_NETWORK_STATE, FOREGROUND_SERVICE (+ FOREGROUND_SERVICE_DATA_SYNC), POST_NOTIFICATIONS, RECEIVE_BOOT_COMPLETED, and REQUEST_IGNORE_BATTERY_OPTIMIZATIONS.

7 OEM & Power-Management Notes

Some manufacturers (Xiaomi/Redmi, Huawei/Honor, Oppo/Realme, Vivo, Samsung, Tecno, Infinix) kill background apps aggressively regardless of Android's standard rules. For reliable background sync and reminders on those devices:

1. Allow notifications when prompted on first launch (Android 13+ requires it explicitly).
2. Grant the battery-optimisation exemption — the app shows a one-time system prompt after login. Or set it manually: Settings → Apps → RDHIS2 → Battery → Unrestricted.
3. On the OEMs above, also enable Autostart / Allow background activity in the manufacturer's own battery / security app.

If these are not granted the app still works — sync simply happens the next time the user opens it, and reminders may be delayed — but the field experience is worse.

8 Storage-Growth Model

For capacity planning on a shared or long-lived device. All figures are estimates for a national-scale instance.

Item	One-time	Per year of active entry	Notes
Metadata cache	15–60 MB / user	negligible (delta only)	Re-downloaded fully only on first login or after a wipe.
Captured data values	—	2–5 MB	~10 datasets/month × ~100 values, plus completion registrations.
Local audit trail	—	1–3 MB	One row per value change; never cleared by a metadata sync.
Result / history caches	< 5 MB	< 5 MB	Bounded — overwritten, not appended.
Total per user	~20–70 MB	~5–10 MB / year	Multiply by the number of users on the device.

A device used by 5 users for 3 years: roughly $5 \times (50 \text{ MB} + 3 \times 8 \text{ MB}) \approx 370 \text{ MB}$. Well within a 16 GB device; trivial on 32 GB+.

9 Recommended Device Profiles

Profile	Use case	Suggested spec
Field phone	A single health worker, mobile, one facility	Android 11+, arm64, 3 GB RAM, 32 GB storage, 6" 1080p, 4000 mAh, 4G
Facility tablet	Shared at a facility, multiple users, large disaggregated datasets	Android 11+, arm64, 4 GB RAM, 64 GB storage, 8–10" 1200p, Wi-Fi (+ optional LTE)
Supervisor device	Woreda / zone user, dashboards + spot capture	Android 12+, arm64, 4 GB RAM, 64 GB storage, 6.5"+ or tablet, 4G/LTE
Minimum viable	Budget constraint	Android 7.0, armeabi-v7a, 2 GB RAM, 16 GB storage, 5" 720p, 2500 mAh — expect slower first sync and occasional memory pressure

10 Server-Side Note

The app places a light, read-mostly load on the DHIS2 server. It does not change the server's own hardware requirements. The only server-side configuration it depends on:

- CORS allowing the web dev origin (localhost:3000/3001) — web only.
- Data-value auditing enabled if the server-side audit trail is to appear in the app's cell-history sheet (optional).
- A valid TLS certificate chaining to a public CA (no pinning; self-signed certificates will fail in release builds).

Prepared from the build configuration at the current head, 10 September 2026. Companion to docs/20-hardware-and-deployment.md.